

Business Case: Autonomous Yard Control Plane

Eliminating Demurrage Waste Through AI-Powered Yard Orchestration

Problem Statement

KPL's MBA depot loses millions in demurrage fees annually due to unorganized truck queuing. On a typical day, 15-23 fuel tankers arrive from suppliers (Total Energies, Rubis, Shell and others) with no booking system, no queue management, and no automated routing. Trucks crowd the gate, block each other, and compete to reach loading bays first. Breakdowns cause cascading traffic jams. The average truck turnaround is 8 hours against an optimal target of 4.5 hours. At a demurrage rate of KES 12,000 per hour, each truck wastes KES 42,000 in avoidable charges. This inefficiency also increases fuel waste, driver fatigue, and risk of delays to downstream customers.

Proposed Solution

An AI-powered control plane that automates the entire truck journey. Trucks are booked before departure. At the gate, ANPR scanning verifies the truck against its scheduled manifest and issues a digital token. A smart engine assigns each truck to the optimal gantry bay based on pump rate, queue depth, and product match. Real-time monitoring flags delays and dispatches personnel if a truck stalls. When loading completes on schedule, pre-movement SMS alerts stage the next driver. Continuous analytics refine bay assignments over time, reducing congestion and demurrage with every cycle.

Financial Impact

Metric	Value
Current avg. turnaround per truck	8 hours
Target avg. turnaround per truck	4.5 hours
Hours saved per truck	3.5 hours
Demurrage rate	KES 12,000 / hour
Savings per truck	KES 42,000
Avg. trucks per day	12
Daily savings	KES 504,000
Annual operating days	300
Annual savings	KES 151,200,000
Implementation cost (Year 1)	KES 8,000,000
Software development & integration	KES 5,000,000
Hardware (ANPR cameras, sensors)	KES 2,000,000
Infrastructure & Year 1 support	KES 1,000,000
Payback period	~ 16 days
ROI (Year 1)	1,790%

Math: 3.5 hrs saved x KES 12,000/hr = KES 42,000/truck. 12 trucks/day x KES 42,000 = KES 504,000/day. KES 504,000 x 300 days = KES 151,200,000/year. Payback = KES 8,000,000 / KES 504,000/day = ~16 days.

Risks & Mitigation

Risk 1: Low adoption by drivers and depot operators - resistance to new processes.

Mitigation: Phased rollout starting with a 2-week pilot at one gate. Hands-on training, clear SMS notifications for drivers, and manual override at every stage ensure no disruption to existing operations.

Risk 2: System downtime disrupting live yard operations.

Mitigation: Every digital token can be verified manually at backup terminals. A 99.9% uptime SLA with redundant infrastructure. The control plane degrades gracefully - if the system is down, depot reverts to manual dispatch with no data loss.

Recommendation

GO - Invest KES 8,000,000 for KES 151M+ annual savings.

With a payback period under 3 weeks and a Year 1 ROI of 1,790%, this solution pays for itself almost immediately. The investment is modest, the operational impact is significant, and the competitive advantage of a fully autonomous depot positions KPC as a regional leader in downstream logistics efficiency.